

Server Components and Virtual Server Environment

Task 1

Install VirtualBox or VMware Workstation Player on the computer and create a new virtual machine named “MyServerLab” using the default settings.

Answer:

1. Opened the virtualization software and selected the option to create a new virtual machine.
2. Entered the virtual machine name as “MyServerLab”.
3. Selected the required operating-system type and version.
4. Kept the default settings for the initial virtual machine creation.
5. Created the VM successfully and verified that “MyServerLab” appears in the virtual-machine list.

Result:

The “MyServerLab” virtual machine was created successfully and is ready for hardware configuration.

Task 2

Configure the “MyServerLab” virtual machine with 2 CPU cores, 4 GB RAM, and a 40 GB virtual hard disk.

Answer:

1. Opened the Settings of the “MyServerLab” virtual machine.
2. In the System/Processor section, assigned 2 CPU cores.
3. In the Memory section, assigned 4096 MB (4 GB) RAM.
4. In the Storage section, created/selected a virtual hard disk with a capacity of 40 GB.
5. Saved the settings and verified the configured values before starting the VM.

Configuration:

- CPU: 2 cores
- RAM: 4 GB
- Virtual Hard Disk: 40 GB

Result:

The virtual server hardware was configured with the required CPU, memory, and storage resources.

Task 3

Add a second virtual network interface card (NIC) to “MyServerLab” and configure one NIC as NAT and the other as Bridged mode.

Answer:

1. Opened the network settings of “MyServerLab”.
2. Enabled the first network adapter and selected NAT mode.
3. Enabled a second network adapter.
4. Selected Bridged Adapter mode for the second NIC and chose the physical network interface used by the computer.
5. Saved the configuration and verified that two network adapters are available.

Network configuration:

- NIC 1 → NAT → Provides internet access through the host system.
- NIC 2 → Bridged Adapter → Connects the VM directly to the physical LAN.

Simple layout:

MyServerLab

■■■ NIC 1 → NAT → Host Computer → Internet

■■■ NIC 2 → Bridged → Physical LAN / Router

Result:

The virtual server has two network connections, allowing it to communicate through NAT and directly with the local network through Bridged mode.

Task 4

Boot the virtual machine and use a system-information tool to identify the CPU, RAM, storage, and network interfaces.

Answer:

After starting “MyServerLab”, system information can be checked using System Information on Windows or “lshw” on Linux. The expected hardware information is recorded below.

System information:

- CPU: 2 virtual CPU cores
- RAM: Approximately 4 GB
- Storage: 40 GB virtual disk
- Network Interface 1: NAT adapter
- Network Interface 2: Bridged adapter

Linux verification command:

`lshw -short`

Windows verification:

Win + R → msinfo32 → System Summary

Result:

The virtual machine detects the CPU, memory, storage, and both configured network interfaces.

Task 5

Change the RAM allocation of “MyServerLab” from 4 GB to 2 GB, restart the virtual machine, and observe the effect on performance.

Answer:

1. Shut down the virtual machine safely.
2. Open the VM Settings and reduce the memory allocation from 4096 MB to 2048 MB.
3. Start the virtual machine again.
4. Observe the startup time and general responsiveness.
5. With less RAM available, applications and services may become slower, especially when several programs run at the same time.

Observation:

Reducing RAM from 4 GB to 2 GB leaves fewer memory resources for the operating system and applications. This can increase the use of virtual memory/pagefile and may cause slower performance. A server with more RAM can handle more services and simultaneous workloads efficiently.

Result:

The VM can operate with 2 GB RAM, but 4 GB provides better performance and more resources for server applications.